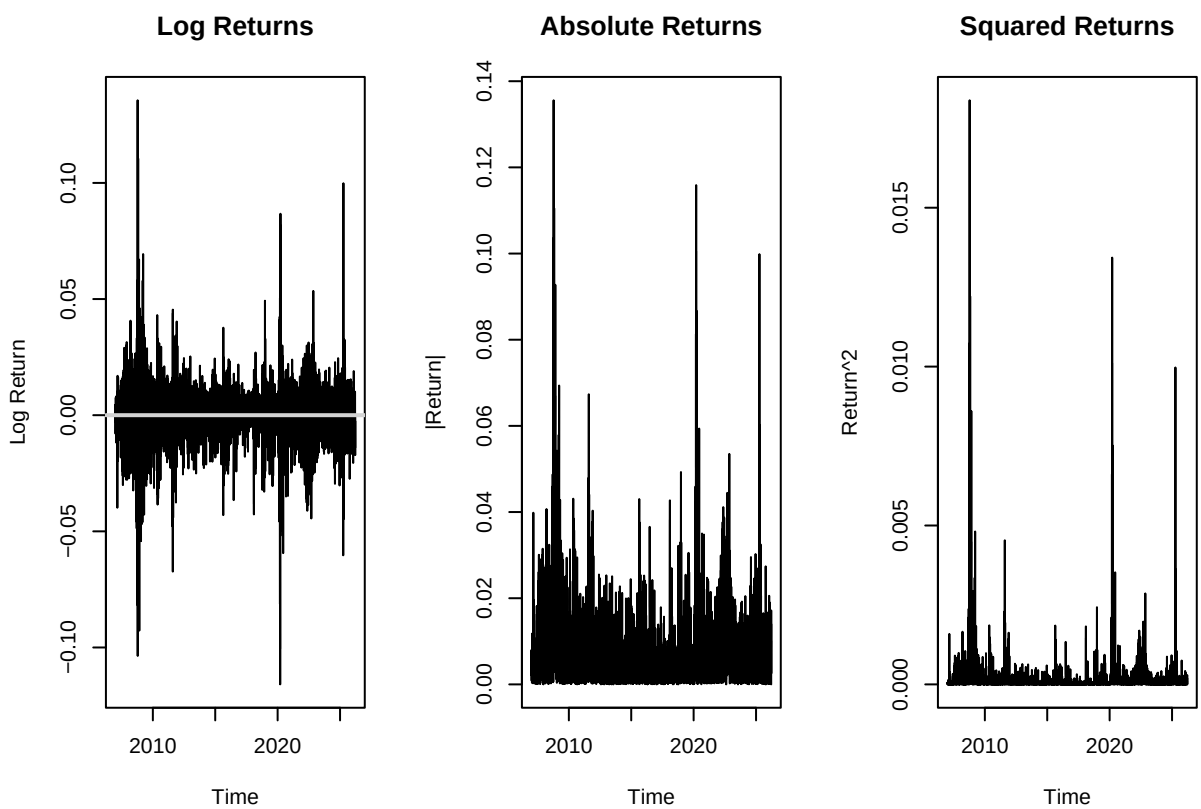
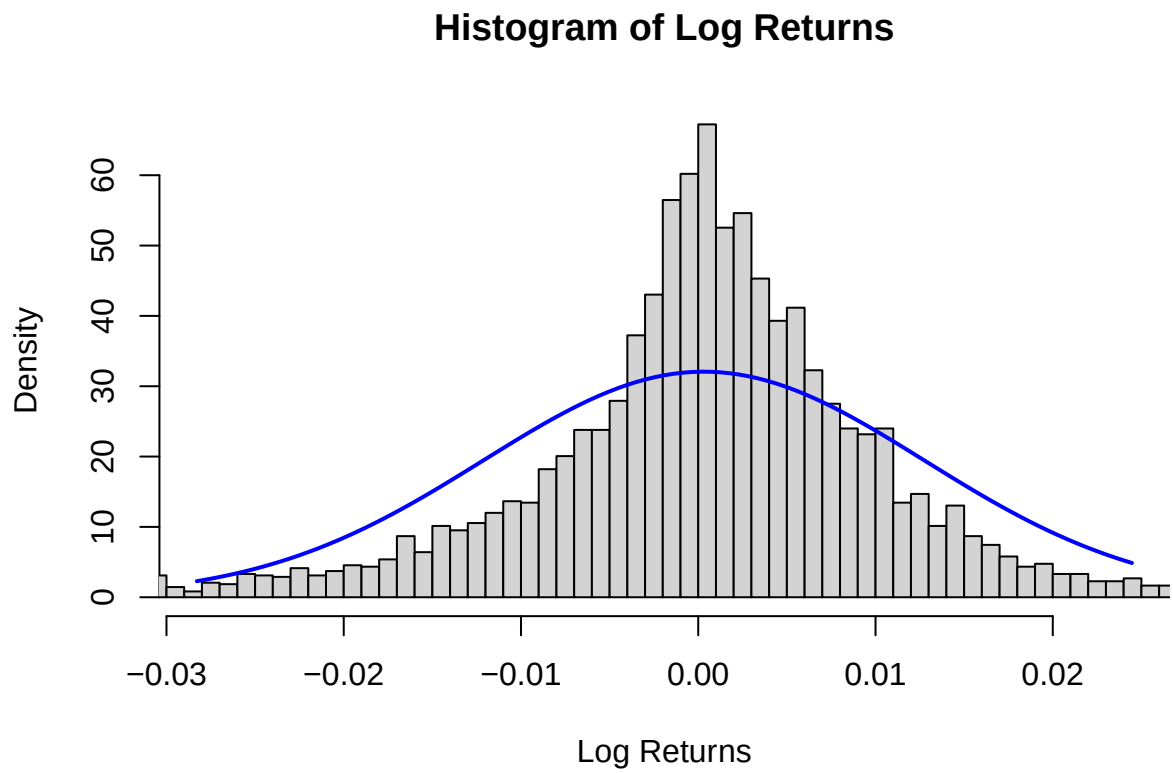
Bayesian Stochastic Volatility Modeling

Data

```
## Loading required package: xts

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

## Loading required package: TTR

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

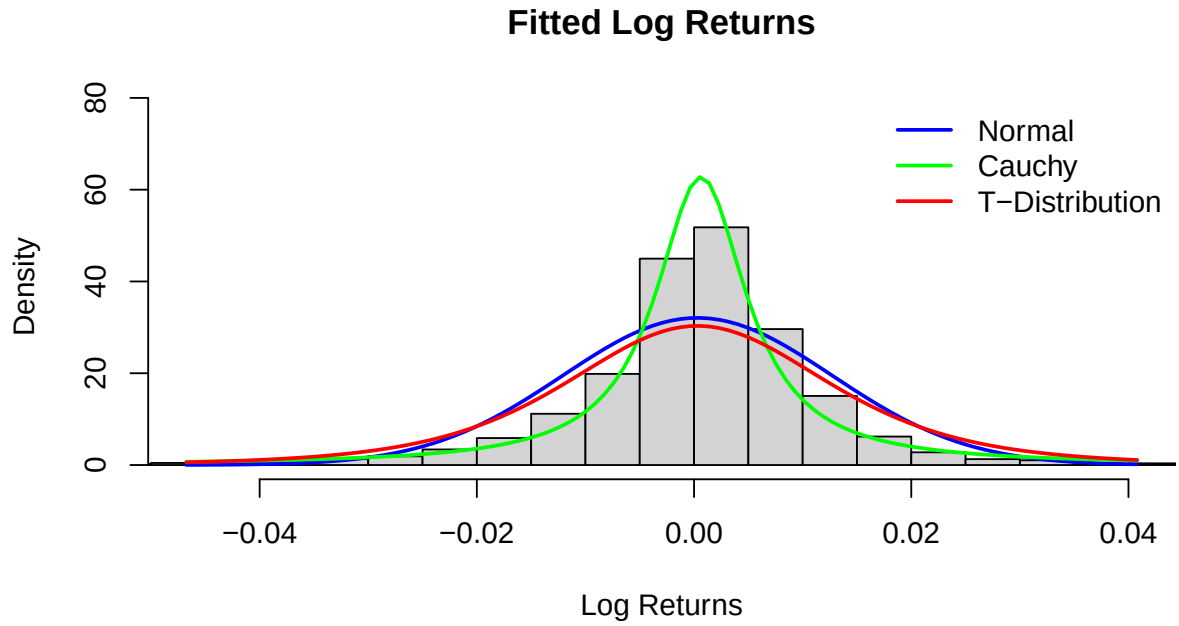


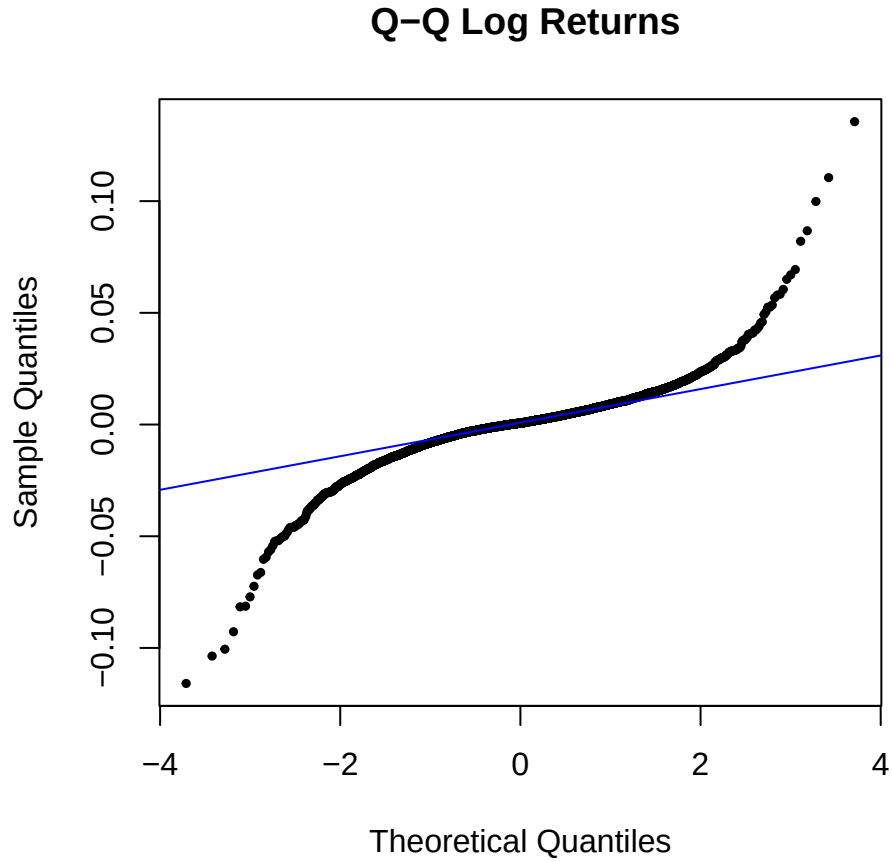


Traditional Models

```
##            model           mu       sigma log_likelihood
## 1         Normal 0.0003151405 0.012440156       14347.27
## 2  Cauchy Aproxx 0.0006329701 0.005069814       14945.98
## 3 T-Distribution 0.0003151405 0.012440156       14659.04
```

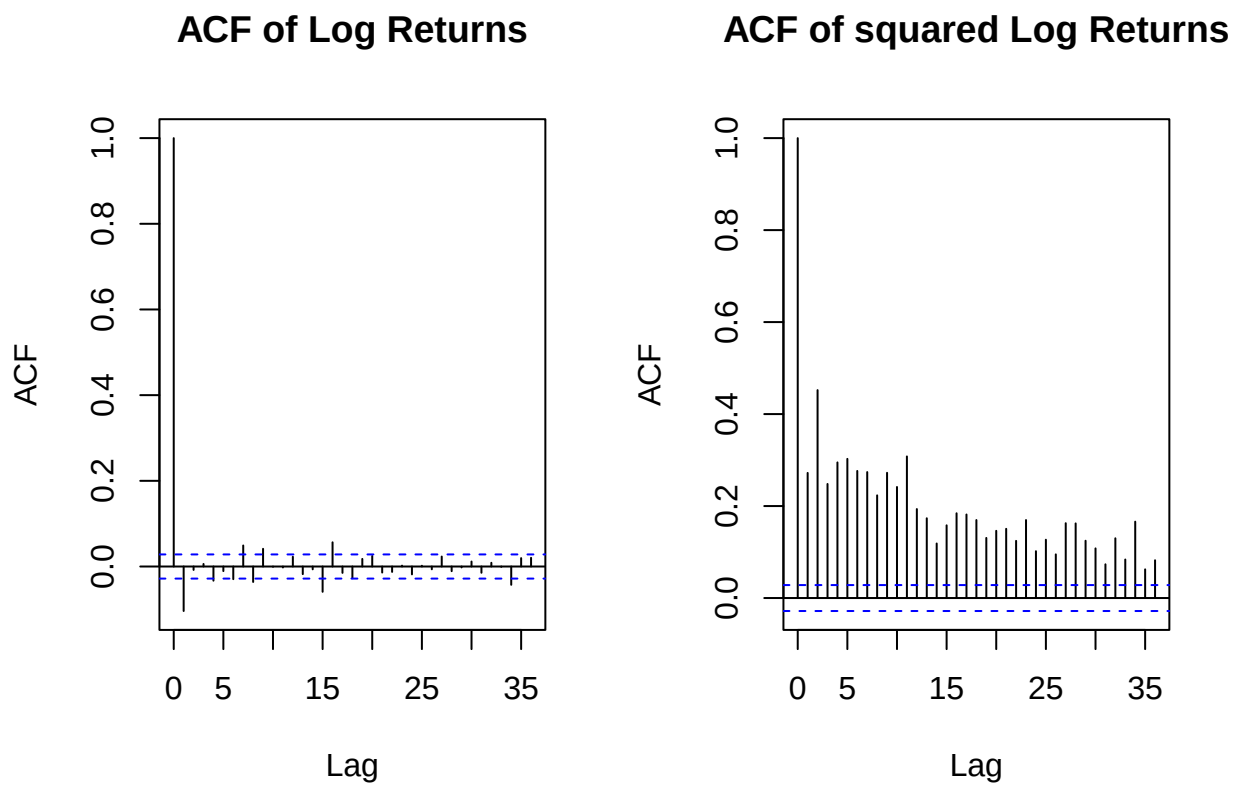
```
## Warning in hist.default(log_return, breaks = 60, probability = TRUE, plot =
## FALSE): argument 'probability' is not made use of
```


Fitted Log Returns
Normal
Cauchy
T−Distribution
Density
0
20
40
60
80
−0.04
−0.02
0.00
0.02
0.04
Log Returns

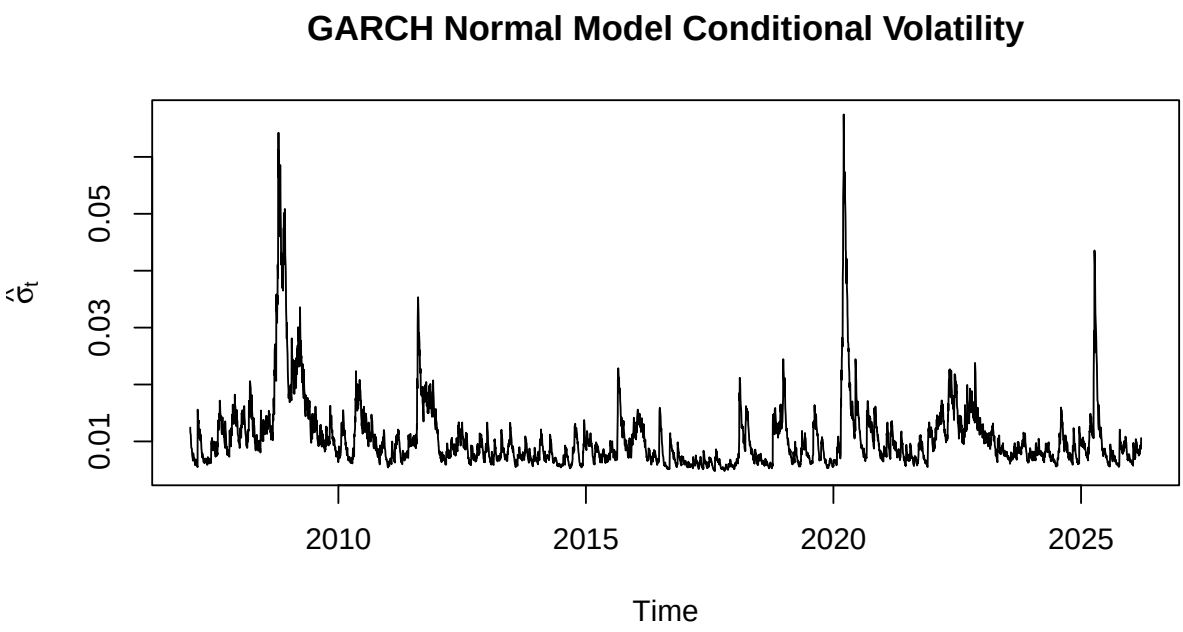

Q−Q Log Returns
Sample Quantiles
−0.10
−0.05
0.00
0.05
0.10
−4
−2
0
2
4
Theoretical Quantiles

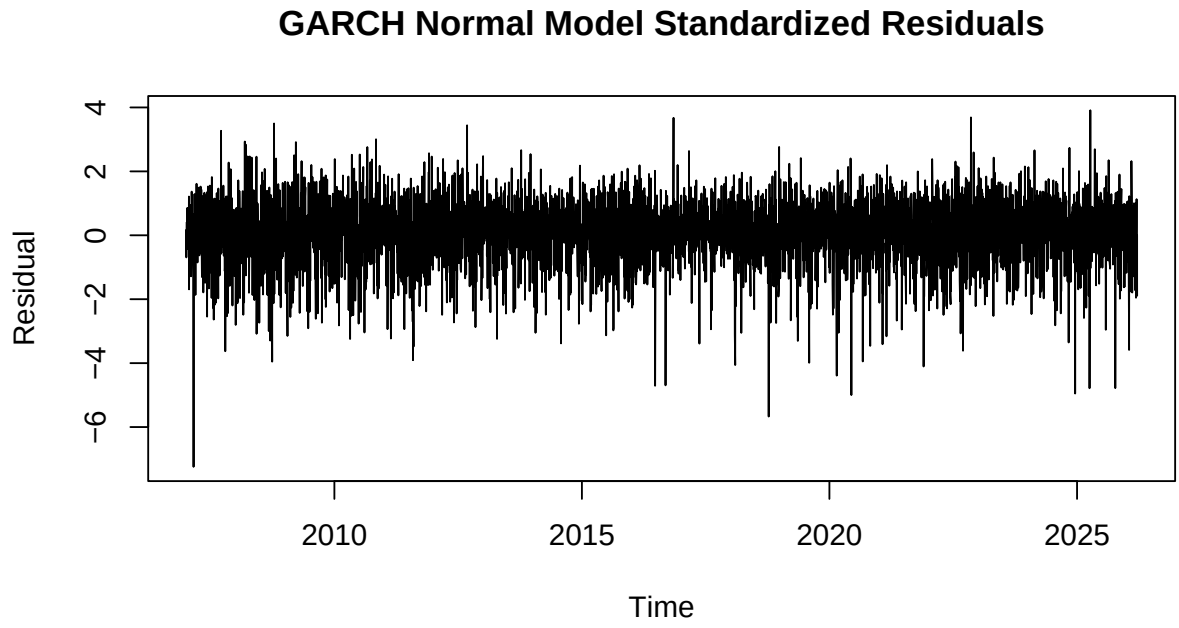
Garch Model

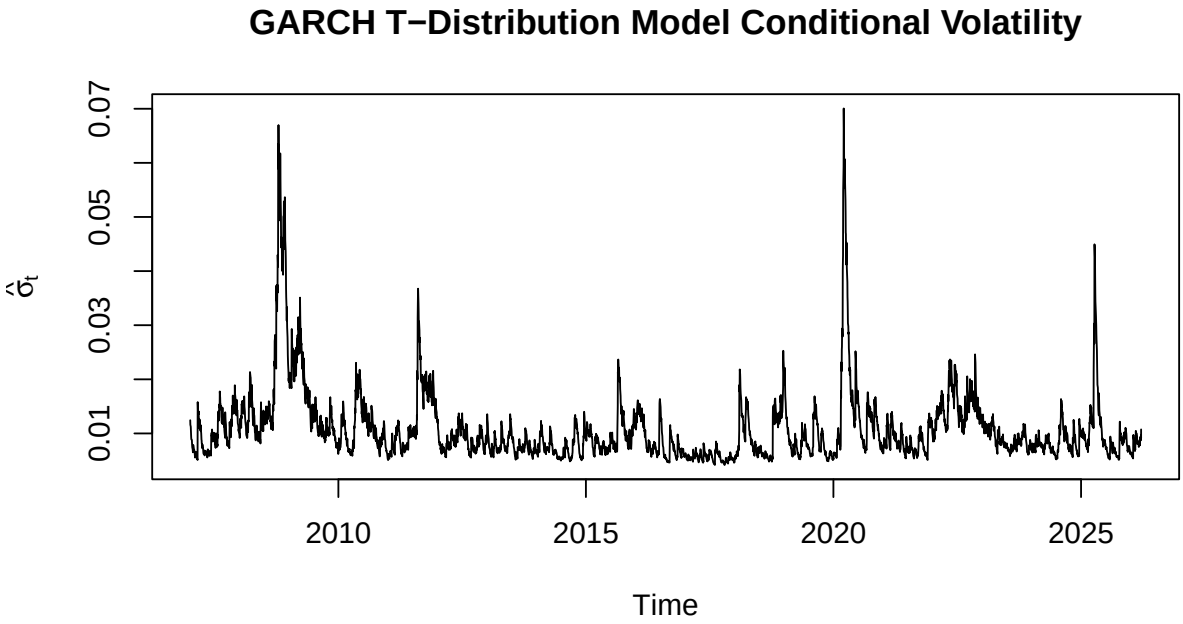
```
## Loading required package: parallel
```

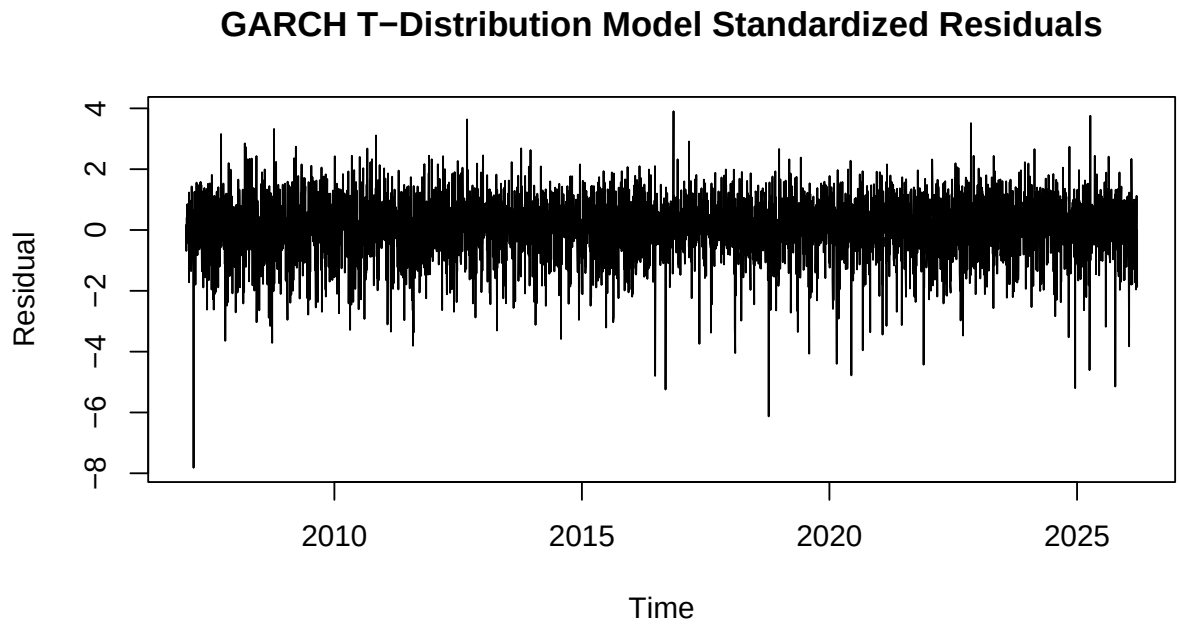



```
##                              model log_likelihood n_parameters        omega
## omega          GARCH Normal Model       15635.94            3 3.204495e-06
## omega1 GARCH T-Distribution Model       15771.61            4 2.020749e-06
##           alpha1     beta1
## omega  0.1349141 0.8408016
## omega1 0.1419426 0.8528575
```


GARCH Normal Model Conditional Volatility
$\hat{\sigma}_t$
0.01
0.03
0.05
2010
2015
2020
2025
Time


GARCH Normal Model Standardized Residuals
Residual
−6
−4
−2
0
2
4
2010
2015
2020
2025
Time





Bayesian Stochastic Volatility Model

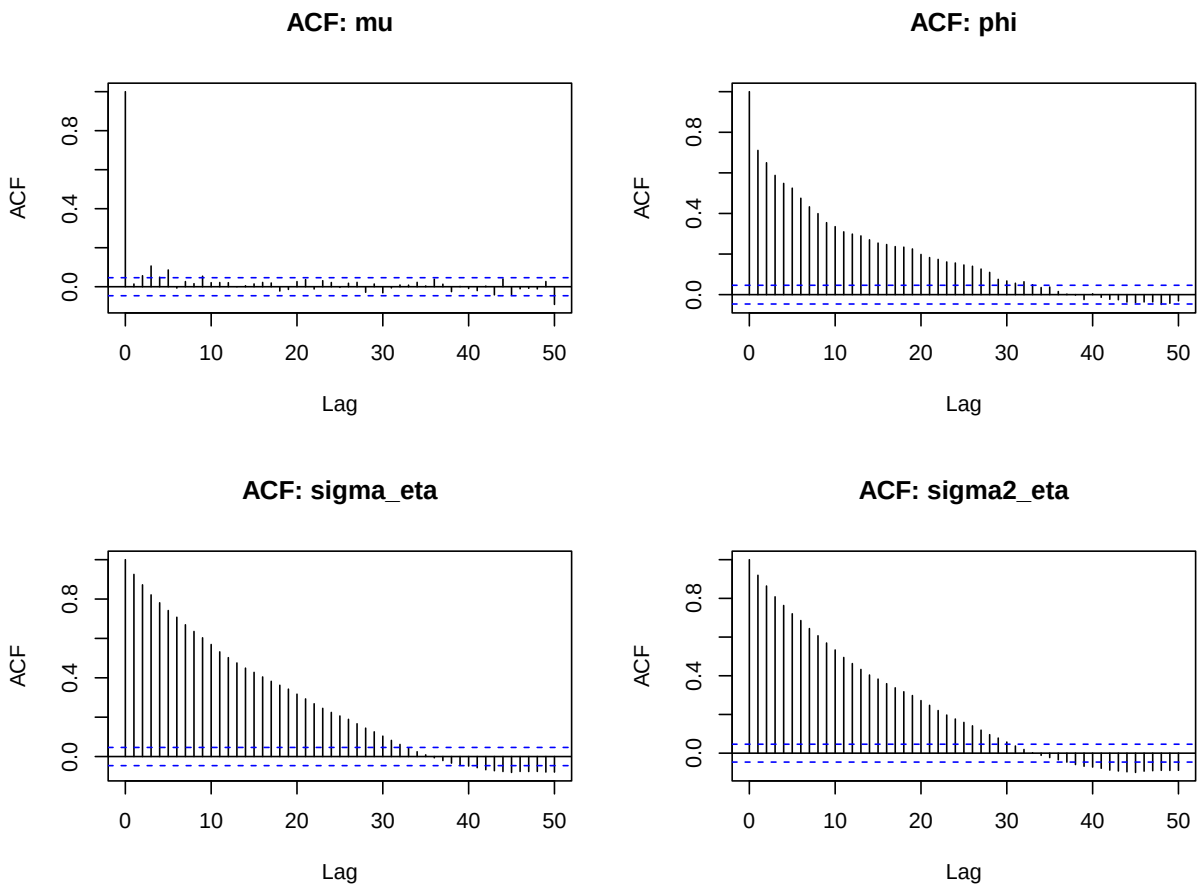
Bayesian Stochastic Volatility Model

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.2

## Loaded modules: basemod,bugs

## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 500
##    Unobserved stochastic nodes: 503
##    Total graph size: 3511
##
## Initializing model

## [1] "model"   "samples" "data"

## List of 3
##  $ model  :List of 8
##   ..- attr(*, "class")= chr "jags"
##  $ samples:List of 3
##   ..- attr(*, "class")= chr "mcmc.list"
##  $ data   :List of 2

## [1] 500

## [1] -0.0027666301 -0.0018487208  0.0083687036 -0.0001911141 -0.0017413094
## [6] -0.0063784007
```

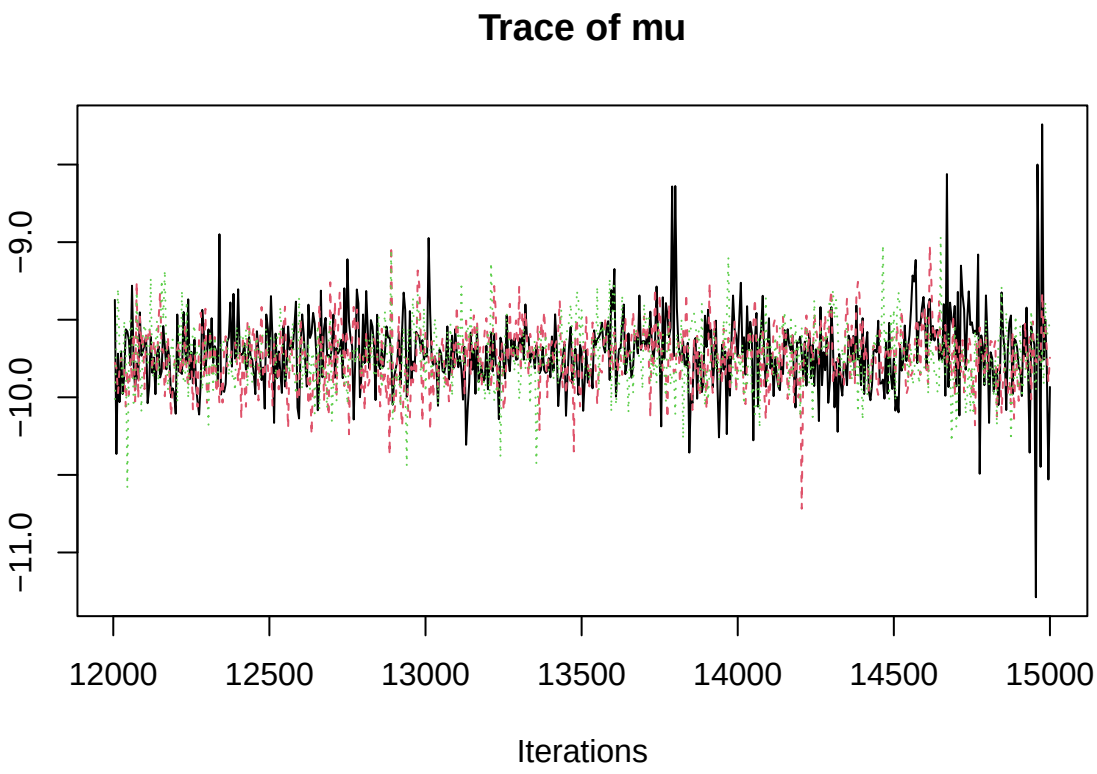
MCMC Diagnostics

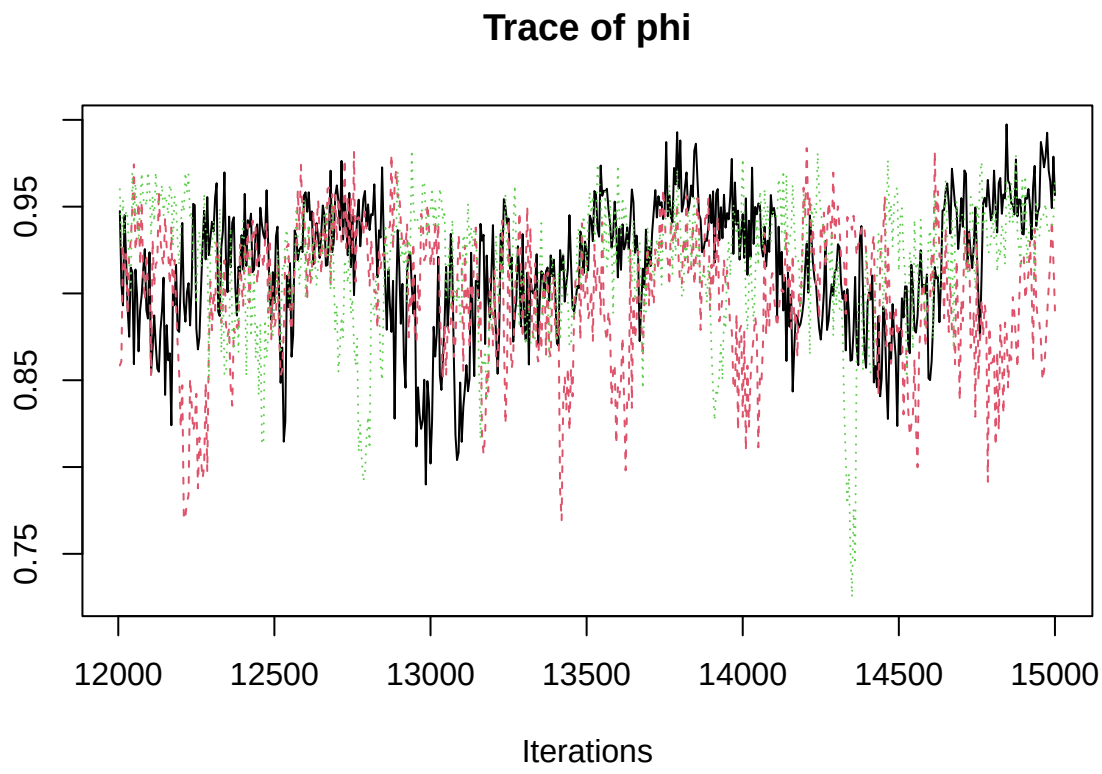
```
##                  Mean         SD     Naive SE Time-series SE
## mu         -9.7273018 0.22217274 0.0052366618    0.006522240
## phi         0.9106476 0.03856048 0.0009088793    0.003891813
## sigma_eta   0.3536011 0.07322875 0.0017260183    0.010150468
## sigma2_eta  0.1303932 0.05456797 0.0012861794    0.006990162

##                    2.5%         25%        50%        75%      97.5%
## mu         -10.13840163 -9.85848989 -9.7231482 -9.6016777 -9.3046867
## phi          0.81820797  0.88775268  0.9163826  0.9392057  0.9699084
## sigma_eta    0.22685819  0.29950590  0.3490043  0.4010966  0.5159140
## sigma2_eta   0.05146464  0.08970387  0.1218040  0.1608785  0.2661672

##                  Mean         SD     Naive SE Time-series SE         2.5%
## mu         -9.7273018 0.22217274 0.0052366618    0.006522240 -10.13840163
## phi         0.9106476 0.03856048 0.0009088793    0.003891813   0.81820797
## sigma_eta   0.3536011 0.07322875 0.0017260183    0.010150468   0.22685819
## sigma2_eta  0.1303932 0.05456797 0.0012861794    0.006990162   0.05146464
##                   50%      97.5%
## mu         -9.7231482 -9.3046867
## phi         0.9163826  0.9699084
## sigma_eta   0.3490043  0.5159140
## sigma2_eta  0.1218040  0.2661672
```

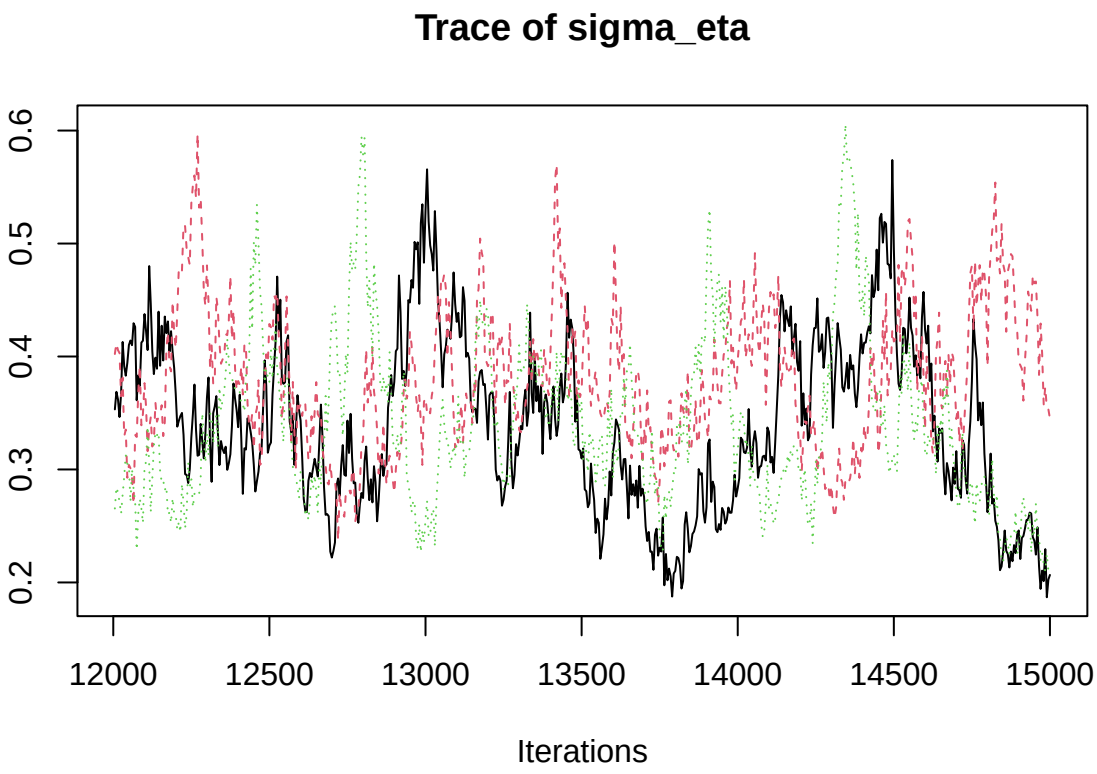


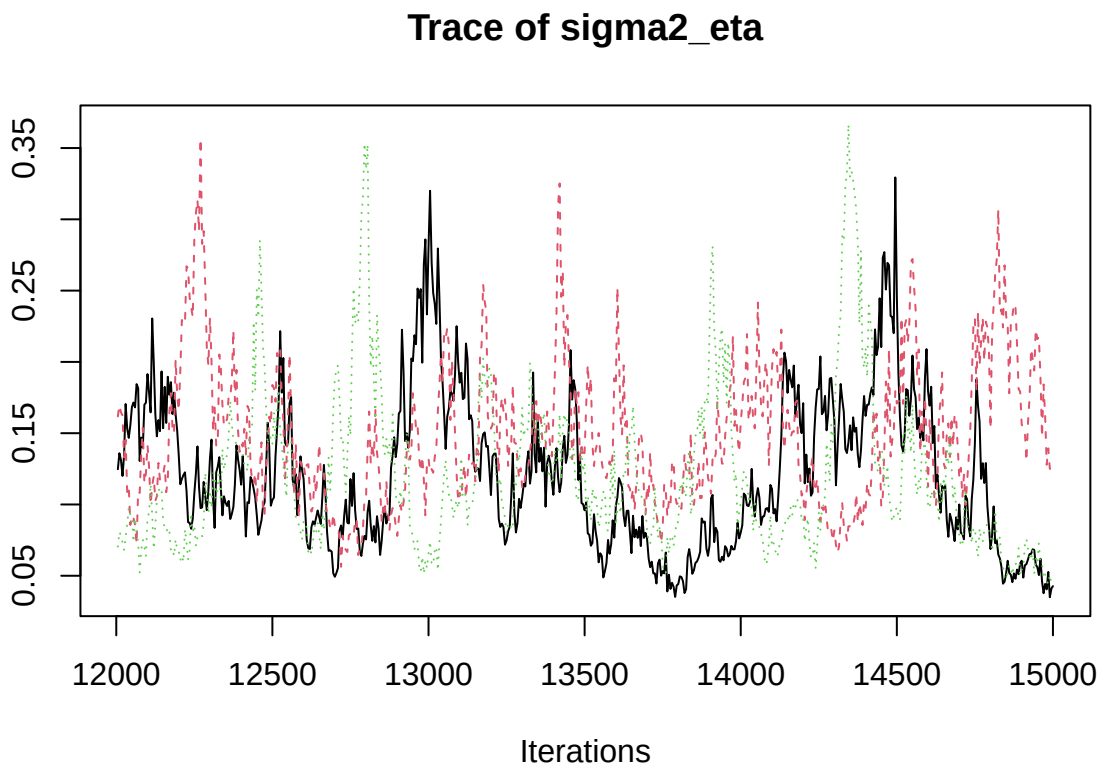

```
## Potential scale reduction factors:
##
##             Point est. Upper C.I.
## mu               1.01       1.02
## phi              1.05       1.17
## sigma_eta        1.08       1.26
## sigma2_eta       1.06       1.21
##
## Multivariate psrf
##
## 1.09
```

```
##         mu        phi  sigma_eta sigma2_eta
## 1229.47090  101.02677   54.82836   61.43178
```

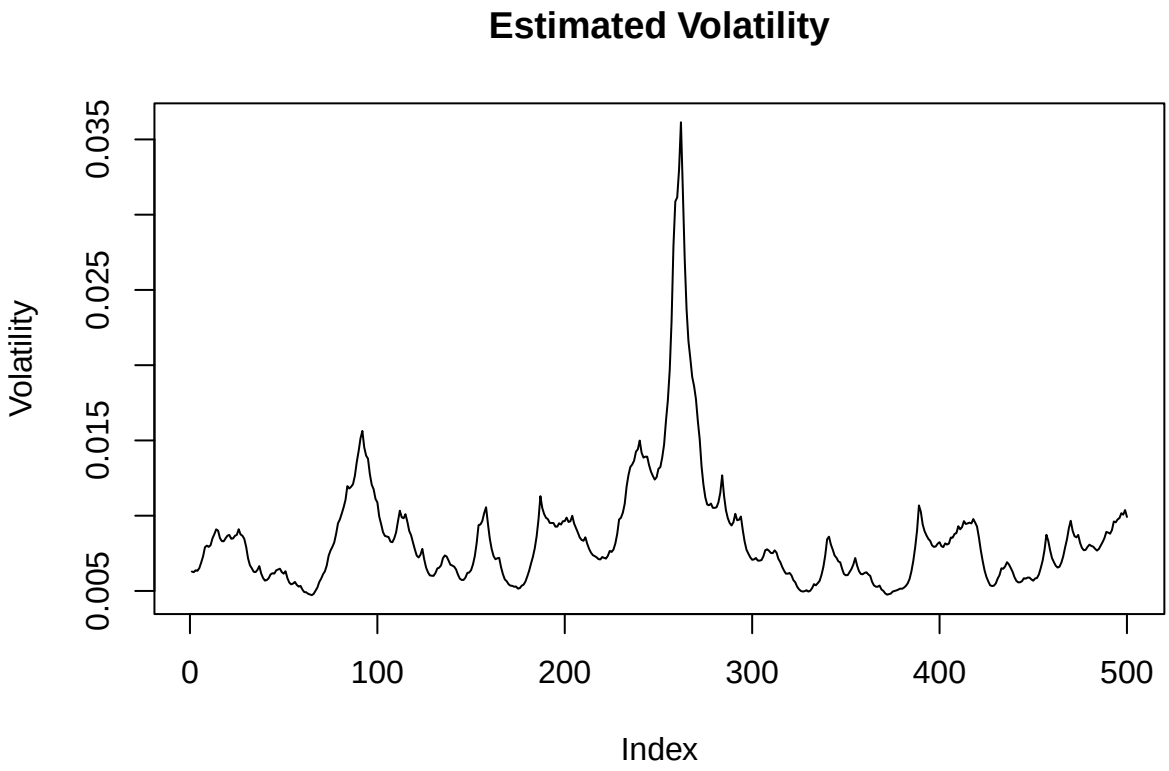

Trace of mu
−9.0
−10.0
−11.0
12000
12500
13000
13500
14000
14500
15000
Iterations

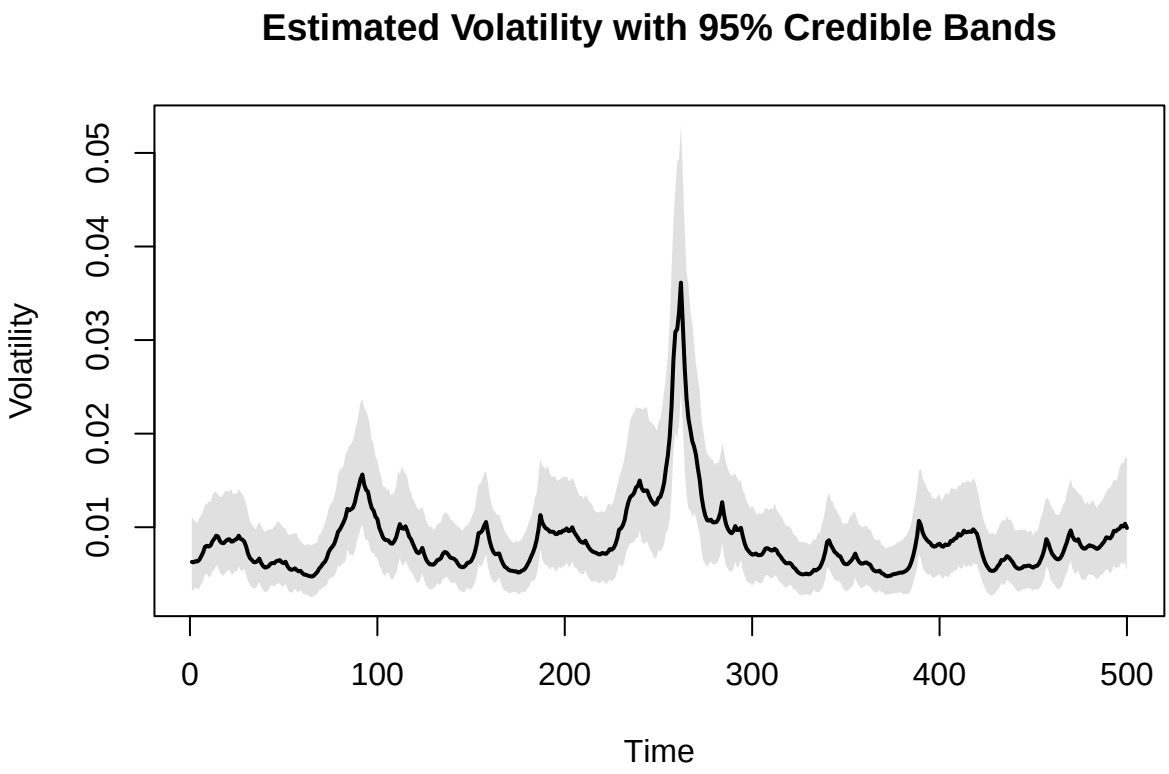

Trace of phi
0.75
0.85
0.95
12000
12500
13000
13500
14000
14500
15000
Iterations

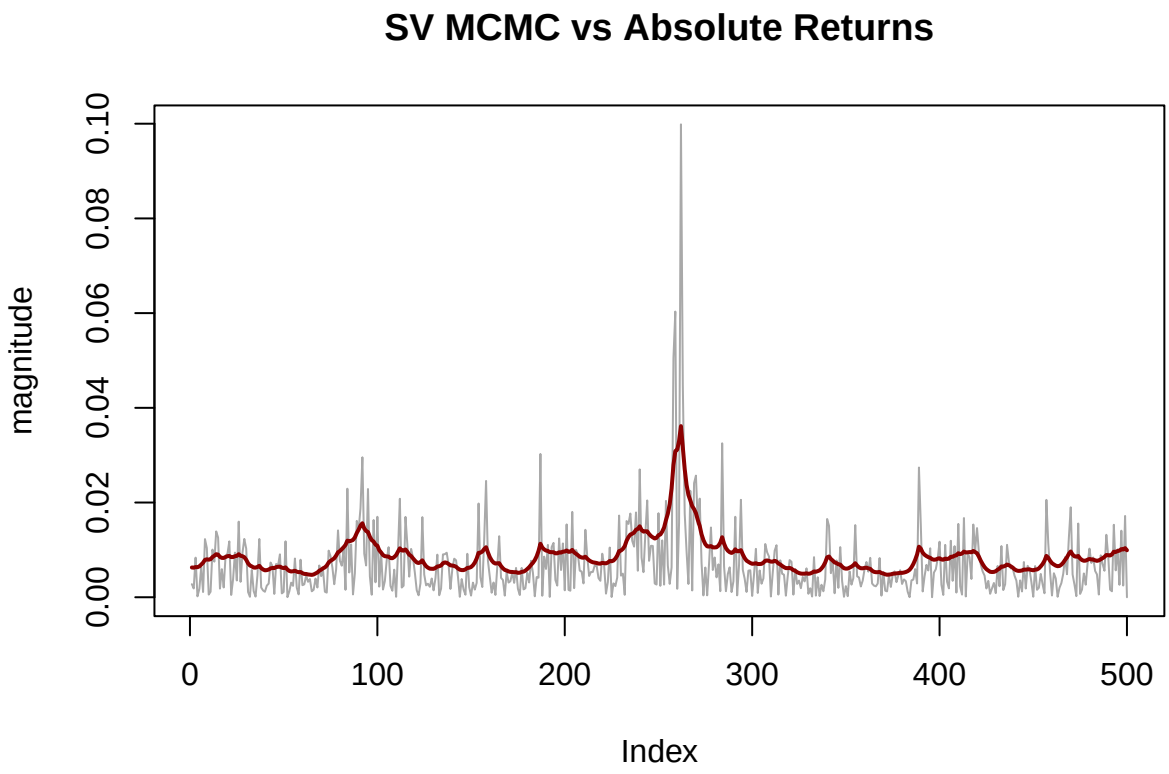

Trace of sigma_eta
0.2
0.3
0.4
0.5
0.6
12000
12500
13000
13500
14000
14500
15000
Iterations

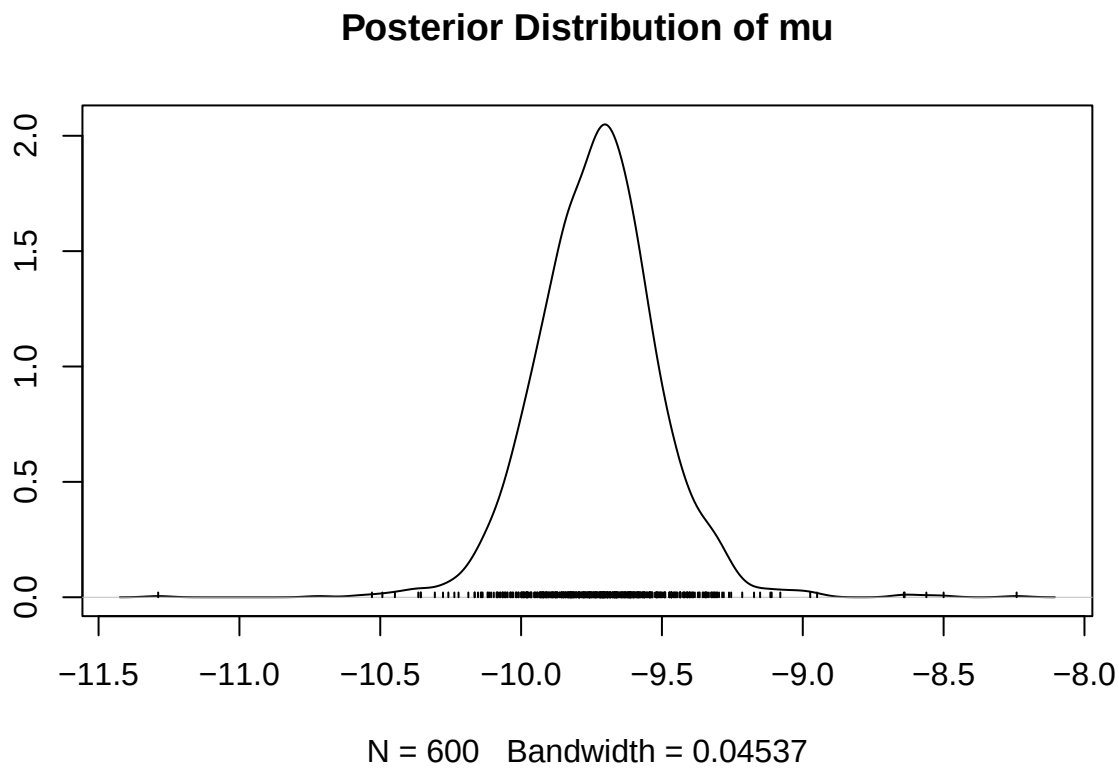

Trace of sigma2_eta
0.05
0.15
0.25
0.35
12000
12500
13000
13500
14000
14500
15000
Iterations

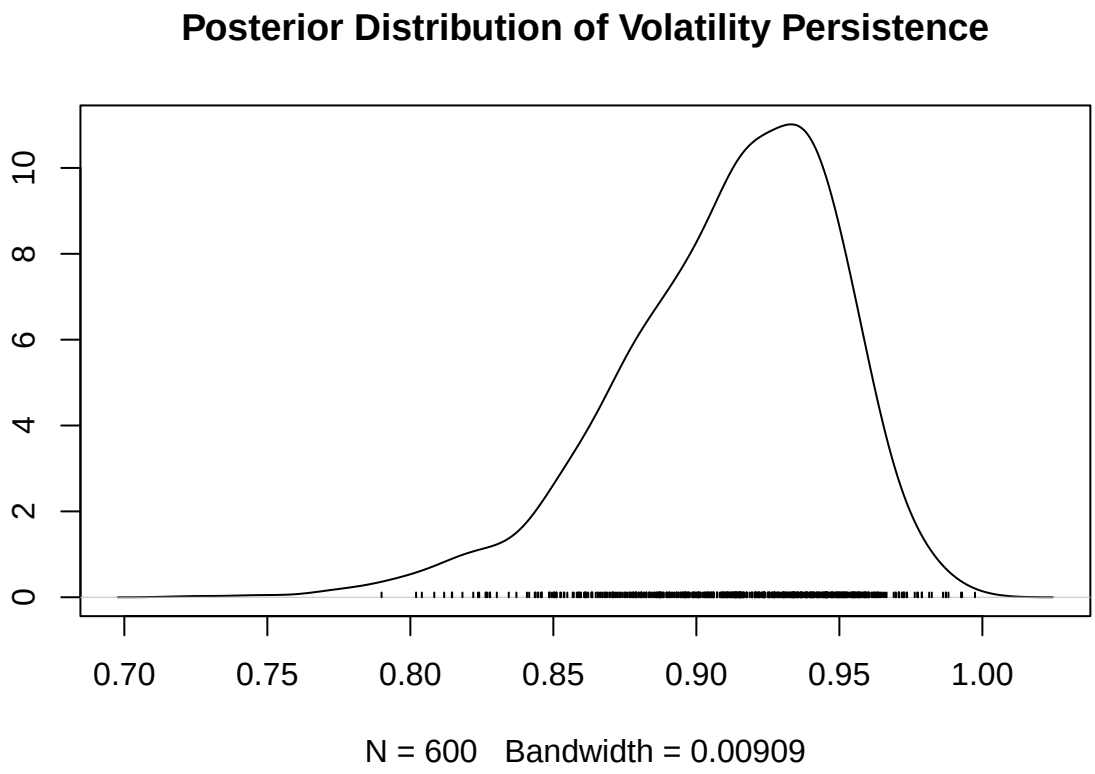
SV MCMC Analysis

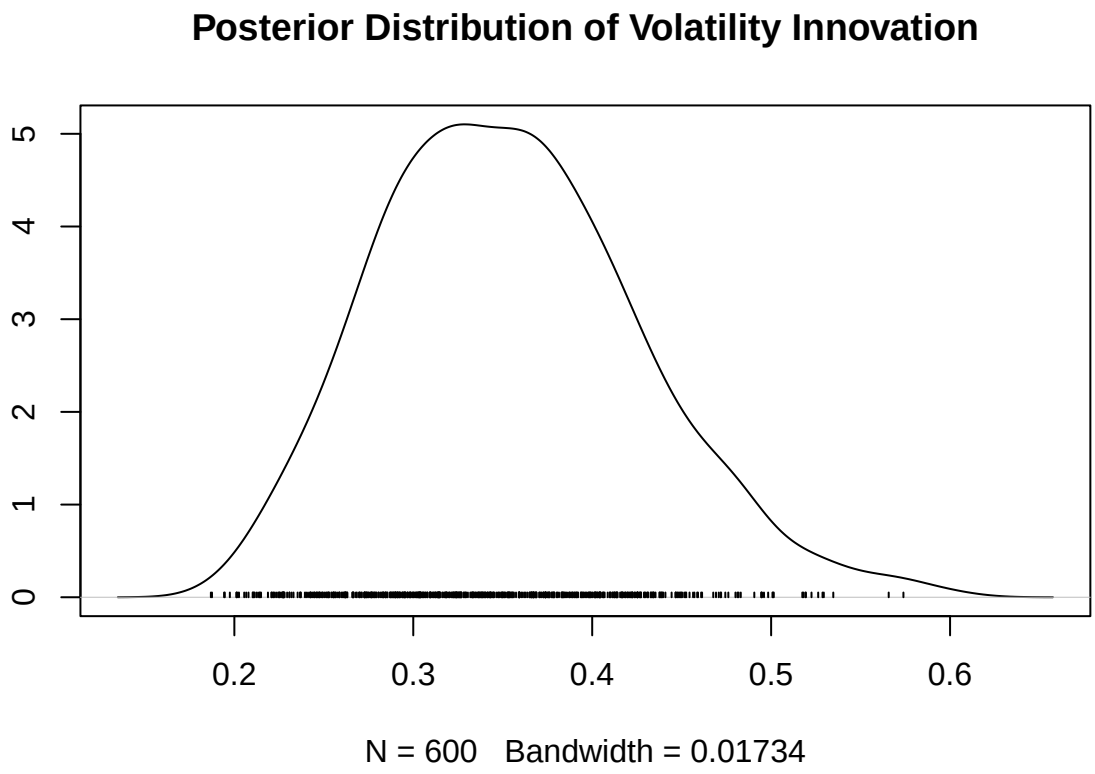


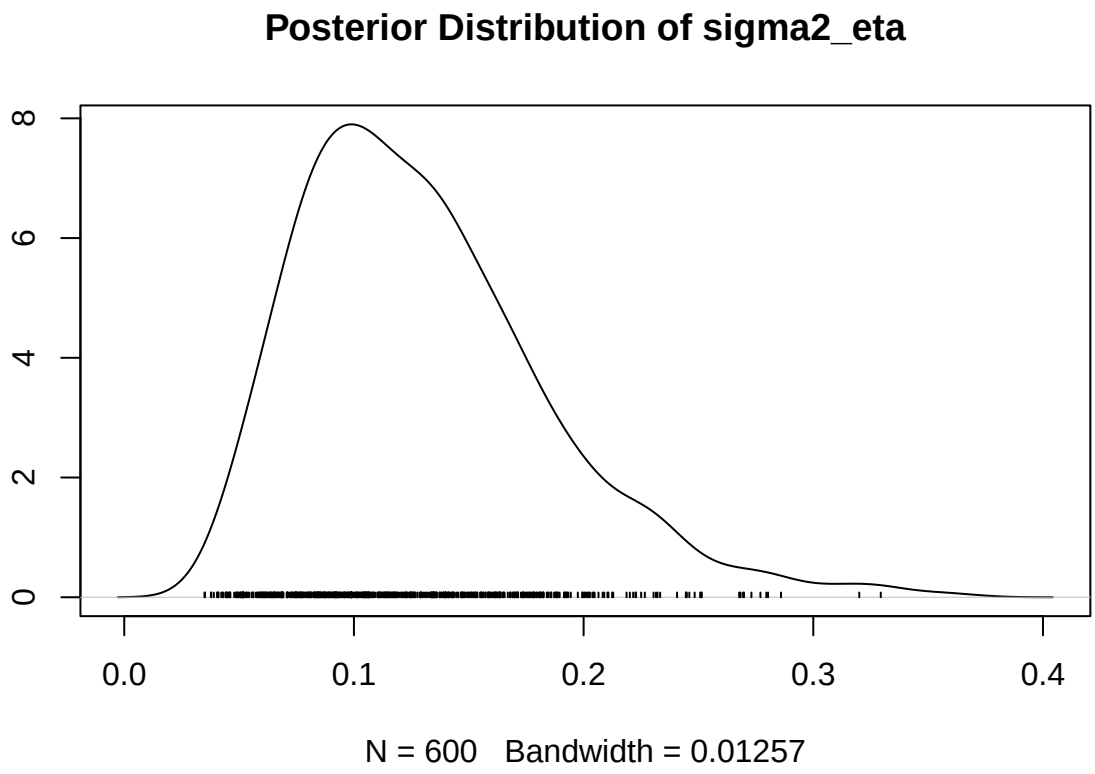

Estimated Volatility with 95% Credible Bands
Volatility
0.01
0.02
0.03
0.04
0.05
0
100
200
300
400
500
Time


SV MCMC vs Absolute Returns
magnitude
0.00
0.02
0.04
0.06
0.08
0.10
0
100
200
300
400
500
Index


Posterior Distribution of mu
0.0
0.5
1.0
1.5
2.0
−11.5
−11.0
−10.5
−10.0
−9.5
−9.0
−8.5
−8.0
N = 600 Bandwidth = 0.04537


Posterior Distribution of Volatility Persistence
0
2
4
6
8
10
0.70
0.75
0.80
0.85
0.90
0.95
1.00
N = 600 Bandwidth = 0.00909


Posterior Distribution of Volatility Innovation
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
N = 600 Bandwidth = 0.01734



Forecasting